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**THE UNIVERSITY OF
BAMENDA
COLLEGE OF
TECHNOLOGY**
*« Building capacities in innovative
technology for development »*
**DEPARTMENT OF
COMPUTER ENGINEERING**

THE UNIVERSITY OF BAMENDA

A Dissertation Submitted to the Department of Computer Engineering in the College of Technology of The University of Bamenda in Partial Fulfillment of the Requirements for the Award of a Master Degree in Computer Engineering.

BY:

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PROF ONABID MATHIAS
(Associate Professor)

AUGUST 2026



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DECLARATION OF ORIGINALITY OF STUDY

I, NJITAPON AHMED SAID ASSAN, registration N° : UBA2SPP228, in the Department of Computer Engineering, College of Technology, The University of Bamenda hereby declare that, this work titled “THE UNIVERSITY OF BAMENDA” is my original work. It has not been presented in any application for a degree or any academic pursuit. I have acknowledged all borrowed ideas nationally and internationally through citations.

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CERTIFICATION OF CORRECTIONS AFTER DEFENSE

This is to certify that this dissertation titled “THE UNIVERSITY OF BAMENDA” is the original work of NJITAPON AHMED SAID ASSAN. This work is submitted in partial fulfillment of the requirements for the award of a Master Degree in Computer Engineering in the College of Technology of The University of Bamenda, Cameroon.

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ABSTRACT

This study investigates and presents the implementation of THE UNIVERSITY OF BAMENDA developed in the Department of Computer Engineering at the College of Technology, The University of Bamenda. The primary aim is to resolve operational challenges through modern technological frameworks and rigorous system architecture. The results demonstrate high reliability, performance efficiency, and adherence to academic and industrial standards.

Keywords: *Architecture, System Design, University of Bamenda, COLTECH, Automation, Optimization.*

RÉSUMÉ

« THE UNIVERSITY OF BAMENDA »

Cette étude examine et présente le développement de THE UNIVERSITY OF BAMENDA au sein du Département de Computer Engineering au Collège de Technologie de l'Université de Bamenda. L'objectif principal est de résoudre les défis opérationnels grâce à des cadres technologiques modernes et une architecture système rigoureuse. Les résultats démontrent une grande fiabilité et une conformité rigoureuse aux normes académiques et industrielles.

Mots-clés : *Architecture, Conception Système, Université de Bamenda, COLTECH, Automatisation, Optimisation.*

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DEDICATION

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CHAPTER 1: INTRODUCTION

1.1 Introduction

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1.1 Background

Phishing and social engineering remain among the most persistent and financially damaging classes of cyberattack, exploiting human trust and judgement rather than purely technical vulnerabilities. Attackers routinely impersonate trusted organisations

through email, text messages (smishing), voice calls (vishing) and, increasingly, synthetic media such as deepfake audio and video, in order to induce victims to disclose credentials, transfer funds or install malicious software. Industry reporting consistently identifies phishing as the most frequently used initial attack vector in data breaches, and business email compromise alone has been linked to tens of billions of dollars in reported losses over the past decade. The last few years have seen a marked shift in the sophistication of these attacks, driven by the same advances in artificial intelligence (AI) that are used defensively. Generative language models can now produce grammatically flawless, contextually convincing phishing emails at scale, while voice-cloning and video-synthesis tools lower the barrier for convincing impersonation attacks against employees, executives and customers. At the same time, AI and machine learning (ML) have proven effective on the defensive side, enabling systems that classify malicious URLs, detect anomalous email content, and flag suspicious behavioural patterns with a level of speed and consistency that manual or purely rule-based approaches cannot match. Despite this progress, most existing AI-based defences address a single, narrow slice of the problem — for example, URL reputation scoring or spam-style email filtering — implemented as isolated tools rather than as part of a coherent system architecture. Organisations, particularly small and medium-sized ones in resource-constrained environments such as Cameroon, are left to assemble ad-hoc combinations of point solutions, with little unified visibility across email, web, and voice/social channels, and with limited capability to detect the broader social-engineering tactics (pretexting, baiting, authority impersonation) that increasingly accompany AI-generated phishing content. This proposal is concerned with the design of a unified, AI-powered system architecture that brings detection, analysis and user awareness into a single platform.

1.2 Description of the Research Problem

Existing phishing-detection tools are largely single-purpose: a spam filter analyses email headers and text, a URL scanner checks domain reputation, and security-awareness training is delivered separately as a human-resources exercise. These components rarely share data or a common decision layer, which creates blind spots — an attack that combines a legitimate-looking email with a spoofed voice call, for instance, is unlikely to be caught by any single existing tool. Furthermore, the rise of AI-generated phishing content undermines detection approaches that rely heavily on stylistic or grammatical anomalies, since such anomalies are precisely what generative

models are able to eliminate. There is, therefore, a need for a system architecture that (i) integrates multiple AI/ML detection engines — natural-language content analysis, URL/domain analysis, and behavioural/contextual analysis — into a single, modular platform; (ii) is designed to scale and to be extensible to new channels and attack types; and (iii) closes the loop with users through real-time reporting and awareness feedback rather than treating detection and human training as separate concerns. This proposal addresses that gap by proposing, and partially prototyping, such an architecture.

1.3 Research Questions and Objectives

1.3.1 Research Questions

RQ1: What are the architectural and detection limitations of existing AI-based phishing and social-engineering countermeasures? RQ2: What modular system architecture can integrate multiple AI/ML detection engines and a user-facing awareness component into a single, scalable phishing and social-engineering detection platform? RQ3: To what extent does a prototype implementation of the proposed architecture achieve acceptable detection accuracy, response latency and scalability under test conditions?

1.3.2 Objectives

General Objective: To design an AI-powered system architecture for a unified phishing detection and social-engineering awareness platform, and to validate its core detection components through a working prototype.

Specific Objectives:

To review and critically analyse existing AI-based approaches to phishing and social-engineering detection, and to identify their architectural and functional limitations. To design a modular, layered system architecture comprising data-ingestion, feature-extraction, an ensemble of ML/NLP detection engines, a decision and alerting layer, and a user-facing dashboard/reporting module. To develop a prototype implementing the core detection engine(s) of the proposed architecture using publicly available phishing datasets. To evaluate the prototype's detection performance (accuracy, precision, recall, F1-score) and system-level performance (latency, throughput) against baseline approaches.

1.4 Rationale

1.4.1 Justification and Motivation

The motivation for this study stems from the widening gap between the sophistication of AI-enabled social-engineering attacks and the fragmented, single-purpose nature of most deployed defences, particularly in institutions with limited security budgets. A well-designed, modular architecture that unifies detection across channels can be adopted incrementally and extended over time, offering a more sustainable defensive posture than a collection of disconnected tools.

1.4.2 Significance

The study is significant to (i) organisations seeking a practical reference architecture for consolidating phishing and social-engineering defences; (ii) the academic community, by contributing an evaluated architectural design and prototype to the growing body of work on AI-based cybersecurity systems; and (iii) the candidate, as partial fulfilment of the requirements for the award of the M.Tech in Information Technology and Cybersecurity.

1.5 Scope and Limitation

1.5.1 Scope

The study covers the design of a system architecture for AI-powered phishing and social-engineering detection, spanning email/text content analysis, URL/domain analysis and a user reporting/awareness module. A prototype will be developed to validate the feasibility of the core detection engine(s) using publicly available phishing datasets.

1.5.2 Limitation

The study is limited by the availability of large-scale, labelled datasets covering voice- and video-based social-engineering attacks, which restricts full prototyping of those channels; the prototype will therefore emphasise text- and URL-based detection, with voice/video analysis addressed at the architectural (design) level only. Computing resources available to the candidate may also constrain the scale of model training and load-testing that can be performed.

CHAPTER 2 LITERATURE REVIEW

2.1 Conceptual Overview

Phishing is commonly defined as a form of social engineering in which an attacker impersonates a trustworthy entity, typically through electronic communication, in order to induce a victim to disclose sensitive information or perform a harmful action. Social

engineering is the broader category of manipulation techniques — including pretexting, baiting and impersonation — of which phishing (and its variants, smishing and vishing) is the most common digital expression. Recent scholarship increasingly frames phishing detection within this broader social-engineering context rather than as a purely technical email- or URL-classification problem.

2.2 Traditional and List-Based Detection Approaches

Early phishing defences relied on blacklists and heuristic rule sets — comparing URLs and domains against known-malicious lists, or flagging emails containing certain keywords. These approaches are simple and low-cost but struggle against previously unseen (zero-day) phishing sites and cannot adapt automatically as attackers change tactics, motivating the shift towards machine-learning-based detection.

2.3 Machine Learning and Deep Learning Approaches

A substantial body of recent work applies supervised machine learning to phishing detection, typically using features extracted from URLs, page content, or email headers and bodies, with classifiers such as Random Forest, Support Vector Machines, and gradient-boosted trees reported to achieve strong detection accuracy on benchmark datasets. Alnemari and Alshammari (2023) demonstrate this approach for phishing-domain classification, achieving high accuracy using ensemble learning on domain-level features. Similarly, Divakarla and Chandrasekaran (2023) apply machine-learning algorithms across both email and website features to improve detection robustness. Deep learning methods, including convolutional and recurrent neural networks, have subsequently been applied to capture more complex patterns in email text and page structure, and are generally reported to outperform classical models on larger, more diverse datasets, at the cost of reduced interpretability and higher computational demand. Kavya and Sumathi (2024) provide a comprehensive review of this trajectory, tracing the field from list-based techniques through machine learning to deep-learning and graph-based methods, and highlighting persistent challenges around dataset diversity and adversarial robustness — that is, the vulnerability of trained models to inputs deliberately crafted to evade them.

2.4 AI-Enabled Social Engineering and Synthetic Media

A newer strand of literature examines how generative AI is being used offensively, producing highly convincing phishing text and, increasingly, synthetic voice and video

content used for impersonation. Mirsky and Lee (2021) provide a comprehensive survey of deepfake creation and detection techniques, cataloguing the generative methods behind synthetic audio and video and the corresponding detection strategies, which is directly relevant to the voice/video dimension of social engineering. Complementary work by Schmitt and Flechais (2024) examines how generative AI lowers the barrier to crafting convincing social-engineering content, underscoring the need for defences that are not solely dependent on the stylistic or grammatical cues that earlier detectors relied upon.

2.5 Datasets and System Architectures

Publicly available datasets such as the UCI Phishing Websites dataset, PhishTank feeds, and labelled phishing-email corpora (e.g., the Vrbančič phishing-websites dataset, 2020) remain the most commonly used resources for training and benchmarking detection models. However, the reviewed literature is dominated by model-level contributions (a classifier trained and benchmarked on one dataset) rather than complete system architectures that integrate multiple detection engines, threat-intelligence feeds and user-facing components into a single deployable platform. Where architectures are discussed, they are typically presented at a high level, without detailed treatment of scalability, modularity, or the integration of a human-awareness/reporting loop.

2.6 Research Gap

The reviewed literature confirms that AI/ML techniques for phishing detection are well established at the model level, and that AI-enabled social engineering (including synthetic media) is a recognised and growing threat. What is comparatively under-addressed is the system-architecture layer: a modular, extensible design that combines multiple detection engines across channels with a real-time user-reporting and awareness component, suited to deployment in resource-constrained institutional settings. This proposal addresses that gap.

CHAPTER 3 MATERIALS AND METHODS (PROPOSED METHODOLOGY)

3.1 Research Design

The study will adopt a Design Science Research (DSR) approach, appropriate for research whose principal contribution is an artefact — in this case, a system architecture and an accompanying prototype. The DSR cycle followed will comprise: (i) problem identification and objective definition (Chapter 1); (ii) design and development of the

architecture and prototype; (iii) demonstration of the prototype against representative datasets; and (iv) evaluation against defined performance metrics, with findings feeding back into refinement of the design.

3.2 Proposed System Architecture Overview

The proposed architecture will be organised into five logical layers: Data Ingestion Layer: connectors for email (via IMAP/API), submitted URLs, and user-reported messages, normalising incoming items into a common internal representation. Feature Extraction Layer: NLP-based text feature extraction (e.g., tokenisation, embedding generation) for message content, and lexical/host-based feature extraction (e.g., domain age, WHOIS data, URL structure) for links. Detection Engine Layer: an ensemble of ML/NLP models — a text-classification model for message content and a URL/domain-classification model — whose outputs are combined by a decision-fusion component to produce a single risk score. Decision and Alerting Layer: applies configurable risk thresholds, triggers alerts, and logs decisions for audit and model-retraining purposes. Presentation Layer: a dashboard for administrators (aggregate threat visibility, model performance) and a lightweight reporting/feedback interface for end users, closing the loop between detection and user awareness. This layered, modular design is intended to allow individual detection engines to be replaced or extended (for example, adding a voice/video-analysis engine) without requiring changes to the rest of the platform.

3.3 Data Collection and Datasets

The prototype will be trained and evaluated using publicly available, ethically sourced datasets, including the UCI Phishing Websites dataset, PhishTank-derived URL feeds, the Vrbančič (2020) phishing-websites dataset, and publicly available labelled phishing-email corpora. No proprietary or personally identifiable data will be collected from real users during the prototype phase.

3.4 Tools and Technologies

Programming and modelling: Python, with Scikit-learn for classical ML models and TensorFlow/PyTorch for deep-learning experiments. NLP processing: standard text-preprocessing and embedding libraries (e.g., spaCy, Hugging Face Transformers) for message-content analysis. Service architecture: a lightweight API framework (e.g., FastAPI/Flask) exposing detection engines as independent services, containerised with Docker for portability. Dashboard: a web-based front end for visualising detection

results and collecting user reports. Version control and documentation: Git for source control and Markdown/LaTeX-style documentation of the design artefacts.

3.5 Model Development

Model development will proceed in two parallel tracks corresponding to the two primary detection engines: (i) a text-classification model trained on labelled phishing/legitimate email and message content, using both classical ML baselines (e.g., Random Forest, SVM) and a deep-learning model for comparison; and (ii) a URL/domain-classification model trained on lexical and host-based features. Model selection will be guided by comparative performance on a held-out validation split, informed by the approaches reviewed in Chapter 2.

3.6 Evaluation Metrics

The prototype will be evaluated using standard classification metrics — accuracy, precision, recall and F1-score — computed on a held-out test set, together with system-level metrics of response latency and throughput measured under simulated load. Where feasible, the combined (ensemble) detection pipeline will be compared against each individual engine to quantify the benefit of the proposed multi-engine architecture.

3.7 Ethical Considerations

The study will use only publicly available, de-identified datasets for model training and evaluation. No real phishing campaigns will be conducted against human subjects, and any demonstration or user-reporting feature will be tested using synthetic or consented sample data only, in line with the University's research-ethics expectations and the COLTECH guidance on academic integrity.

3.8 Proposed Work Plan

Table 4.2: Summary Table

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